

Gas Pressure, the Particle Model and Diffusion — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. This topic is marked on wording far more than on arithmetic, so the schemes below quote the phrases that earn each mark and name the ones that do not. Almost every explanation mark lost on these four papers goes one of four ways: writing that the particles “push harder” when a gas is squashed, writing that the particles expand when heated, saying that a vacuum sucks, or giving only one of the two reasons why heating a fixed volume raises the pressure. The note under each answer names the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 Pressure is the force acting perpendicular to a surface, per unit area of that surface; [2]
 pressure = force ÷ area [1]
 Unit: the **pascal (Pa)**, equal to 1 N/m^2 [1]
“Newtons” alone is not a unit of pressure and scores nothing. Accept N/m^2 in place of Pa.
- 2 (a) $240 \div 0.4 = \mathbf{600 \text{ Pa}}$ [1] [4]
 (b) $60 \div 0.02 = \mathbf{3000 \text{ Pa}}$ [1]
 (c) $1500 \div 0.05$ [1] = $\mathbf{30\,000 \text{ Pa}}$ [1]
Dividing area by force gives 0.0017, 0.00033 and 0.000033. If all three answers come out tiny, the whole question has been done upside down — award the method mark in (c) only.
- 3 (a) $200 \times 3 = \mathbf{600 \text{ N}}$ [1] [3]
 (b) $100\,000 \times 0.02 = \mathbf{2000 \text{ N}}$ [1]
 (c) $50\,000 \times 0.4 = \mathbf{20\,000 \text{ N}}$ [1]
- 4 The gas particles are in constant random motion and collide with the walls of the container [1] [2]
 Each collision exerts a small force on the wall; the total force from all the collisions, divided by the area, is the pressure [1]
“The gas pushes on the walls” restates the question and scores nothing. The word collide (or “collisions”) is required for the first mark.
- 5 The pressure **increases**, roughly doubling [1] [3]
 The particles have a shorter distance to travel between the walls, so they hit the walls more often [1]
 giving more **collisions per unit area per second** [1]
“The particles are squashed so they push harder” scores zero for the explanation. Nothing has speeded the particles up, so no collision is harder than before. This is the single most common wrong answer in the topic.
- 6 The pressure **increases** [1] [3]
 The particles gain kinetic energy and move faster, so they hit the walls **more often** per second [1]
 and each collision transfers more momentum, so **each hit is harder** [1]
Two separate reasons are asked for and most students give only the first. Do not award the third mark for a rewording of the second.
- 7 The particles move **randomly in all directions**, so no direction is favoured [1] [2]
 Every wall therefore receives the same number of collisions per unit area per second [1]
Answers that say the pressure is greatest at the bottom have imported the behaviour of a liquid. The weight of the gas is negligible here.

- 8** There are now more particles in the same volume **[1]** [2]
so more collisions with the walls each second, and therefore more collisions per unit area per second **[1]**
Award nothing for “there is more air in it so the pressure is higher” — that is the question restated.
- 9** The mistake: squashing a gas does not make the particles move any faster, so no collision is harder than before; only temperature changes how hard a particle hits **[1]** [3]
Correct explanation: the particles have less distance to travel between the walls, so they collide with the walls more often **[1]**
giving more collisions per unit area per second, which is a higher pressure **[1]**
The first mark is for identifying why “push harder” is wrong, not just for saying that it is.
- 10** (a) $27\text{ }^{\circ}\text{C} = 300\text{ K}$ and $327\text{ }^{\circ}\text{C} = 600\text{ K}$ (add 273) **[1]** [2]
(b) The kelvin temperature has doubled at constant volume, so the pressure **doubles** **[1]**
A student who reasons from Celsius sees 27 going to 327 and concludes the pressure rises twelvefold. Kelvin is not optional here.
- 11** Diffusion is the **net** movement of particles from a region of higher concentration to a region of lower concentration **[1]** [2]
caused by the **random** motion of the particles **[1]**
Both net and random are required, one for each mark. “Particles spreading out” describes the result, not the process.
- 12** Any two of: a higher temperature; lighter particles (smaller M_r); a greater difference in concentration **[1]** [2]
Explanation, e.g. a higher temperature gives the particles more kinetic energy so they move faster on average, and cover ground more quickly **[1]**
“Stir it” and “use a fan” are not accepted. Those replace diffusion with bulk movement of the gas, which is a different process.

Paper B — Medium

30 MARKS

- 1** The atmosphere is a gas, and its particles are in constant random motion **[1]** [3]
They collide with every surface, and the total force of those collisions divided by the area is the pressure **[1]**
At sea level it is about **100 000 Pa** (accept 100 kPa, or 1×10^5 Pa) **[1]**
- 2** (a) force = pressure $\times$ area = $100\,000 \times 0.01$ **[1]** = **1000 N** **[1]** [4]
(b) area = $1.2 \times 0.6 = 0.72\text{ m}^2$, force = $100\,000 \times 0.72 = 72\,000\text{ N}$ **[1]**
(c) The same atmospheric pressure acts on the underside of the desk, so the forces balance and there is no resultant force **[1]**
1000 N is the weight of a 100 kg person. Worth pointing out to a student who has written it down without reacting to it.
- 3** (a) The steam **condenses** back to liquid water **[1]** [4]
(b) The number of gas particles inside collapses, so there are almost no collisions on the inside wall and the pressure inside falls close to zero **[1]**
(c) **Atmospheric** pressure outside is unchanged at about 100 000 Pa **[1]**; with nothing pushing back from inside, the atmosphere pushes the walls inwards and crushes the can **[1]**
Deduct the final mark from any answer containing “the vacuum sucked it in”. A low pressure does not pull; it simply pushes back far more weakly, so the atmospheric push that was there all along is no longer balanced.

- 4 Expanding your chest lowers the pressure of the air inside the straw **[1]** [3]
 Atmospheric pressure still acts on the surface of the drink in the glass **[1]**
 so the atmosphere pushes the liquid up the straw into the region of lower pressure **[1]**
The question forbids the word “suck” on purpose. A student who cannot write the answer without it has not understood that nothing pulls.
- 5 The net movement of particles from a region of higher concentration to a region of lower concentration **[1]** [3]
 The particles are in constant random motion and never stop moving **[1]**
 Where they are crowded, more happen to wander outwards than wander back, so the movements do not cancel **[1]**
The third mark is the one that separates a description from an explanation. It is rarely written and always available.
- 6 (a) **Diffusion** **[1]** [4]
 (b) The room is already full of air particles **[1]**; each vapour particle collides with them constantly and changes direction, so it follows a long random zigzag path and its progress across the room is far slower than its actual speed **[1]**
 (c) Warming the room (or a greater concentration of vapour); the particles gain kinetic energy and move faster on average **[1]**
“Open a window” or “turn on a fan” scores nothing in (c) — both replace diffusion with a draught.
- 7 In a gas the particles are much further apart and moving much faster **[1]** [3]
 so they travel further between collisions and spread more quickly; in a liquid the particles are touching, slower, and constantly obstruct one another **[1]**
 In a solid the particles only vibrate about fixed positions and cannot move past each other at all **[1]**
- 8 The orange colour spreads **upwards** into the top jar until both jars are evenly pale orange **[1]** [3]
 The bromine and air particles are in constant random motion, so they mix by diffusion **[1]**
 It happens even though bromine is much denser than air, because diffusion is driven by random particle motion and not by weight; it is slow because bromine particles are heavy (M_r 160 against about 29 for air) **[1]**
The direction is the whole point of this classic. An answer predicting that the bromine stays at the bottom because it is heavier has missed it.
- 9 (a) $1.5 \times 0.8 = 1.2 \text{ m}^2$ **[1]** [3]
 (b) $100\,000 \times 1.2 = 120\,000 \text{ N}$ **[1]**
 (c) The same atmospheric pressure acts on the other side, so the two forces balance **[1]**
120 000 N is about the weight of twelve tonnes on one window. The balance in (c) is the only reason glazing survives at all, and stating it is worth the mark.

- 1** (a) The volume is divided by 3, so the pressure is multiplied by 3: **300 kPa [1]**; the particles travel a third of the distance between the walls, so they collide with the walls three times as often **[1]**, giving three times as many **collisions per unit area per second [1]** [5]
 (b) The kelvin temperature doubles at constant volume, so the pressure doubles: $300 \times 2 = \mathbf{600\ kPa\ [1]}$
 (c) The syringe stays sealed and rigid — no gas escapes and the volume does not change **[1]**
In (b), starting again from 100 kPa gives 200 kPa. The two changes happen one after the other, so the second acts on the answer to the first. This is the most common way to lose a mark on Paper C.
- 2** (a) The pressure **falls [1]**; the particles lose kinetic energy and move more slowly, so they hit the walls less often and each collision is gentler **[1]** [5]
 (b) The kelvin temperature has halved at constant volume, so the pressure halves: **50 kPa [1]**
 (c) The mistake: particles never change size — a particle is identical at 300 K and at 150 K **[1]**; what changes is their average speed, which lowers both the number of collisions per unit area per second and the force of each one **[1]**
Award the first mark in (a) even if the reason is wrong, but only if the direction is right. Any mention of the particles shrinking, expanding or changing in (c) other than in speed cancels the second mark there.
- 3** (a) Both gases spread along the tube by **diffusion** — net movement from the plug, where they are concentrated, towards the middle, because of their random motion **[1]**; where they meet they react to form solid **ammonium chloride, NH_4Cl [1]** [6]
 (b) $60 - 36 = \mathbf{24\ cm\ [1]}$; ratio $36 : 24 = \mathbf{1.5}$ (accept $3 : 2$) **[1]**
 (c) Both gases travelled for the same time, so ammonia must have moved faster on average **[1]**; ammonia particles have a smaller mass (17 against 36.5), and at the same temperature lighter particles move faster, so they diffuse further **[1]**
 $\sqrt{36.5 \div 17} = 1.47$, which is why the ring lands where it does. That calculation is beyond this syllabus and must not be required, but a student who produces it deserves to be told they are right.
- 4** (a) The ring appears **sooner [1]**; both gases have more kinetic energy and so move faster on average, so both fronts advance more quickly **[1]** [4]
 (b) The position is **unchanged [1]**; both gases are speeded up by the same factor, so the ratio of the two distances, and therefore the meeting point, is the same **[1]**
Part (b) is the discriminator. Most students assume that anything which changes the timing must also move the ring. The reason it does not is that the change applies equally to both gases.
- 5** The air in the can is made of particles in constant random motion, colliding with the walls; the total effect of those collisions, per unit area, is the pressure **[1]** [6]
 Heating gives the particles more kinetic energy, so their average speed increases **[1]**
 They therefore reach the walls more often — more **collisions per unit area per second [1]**
 and each collision transfers more momentum, so each one exerts a greater force on the wall **[1]**
 The volume cannot change, so neither effect is relieved and the pressure keeps rising **[1]**
 When the outward force on the wall exceeds what the metal can withstand, the can bursts **[1]**
Six marks, six separate ideas. Mark strictly: this question exists to reward a student who writes the mechanism out in order rather than asserting that “heat makes pressure”. Award no mark anywhere in this answer for “the particles expand”.

- 6 (a) $250\,000 \times 0.008 = \mathbf{2000\,N}$ [1] [4]
 (b) $100\,000 \times 0.008 = \mathbf{800\,N}$ [1]
 (c) $2000 - 800 = \mathbf{1200\,N}$, acting **outwards** [1]
 (d) **1200 N inwards** [1]

Ignoring the atmosphere altogether and answering 2000 N in (c) is the standard error. Atmospheric pressure acts on the other side of everything and is never decoration in a piston question.

Paper D — Hard

30 MARKS

- 1 (a) The particles gain kinetic energy and move faster, so they hit the walls more often [1] and each collision is harder, giving a greater force on the same area [1] [5]
 (b) Steam escaping removes particles from the pot [1], so there are fewer particles left to collide with the walls and therefore fewer collisions per unit area per second [1]
 (c) The pressure inside is far above atmospheric, so there is a large resultant outward force on the lid, which would otherwise be blown off [1]

Part (b) is the unfamiliar one. Students who have learnt “pressure goes up when you heat” as a rule rather than a mechanism have nothing to say here.

- 2 (a) The higher you go, the less air there is above you [1], so there are fewer particles per unit volume and therefore fewer collisions per unit area per second [1] [5]
 (b) Pressure $\times$ volume stays constant, so $100 \times 2.0 = 25 \times V$ [1], giving $V = \mathbf{8.0\,m^3}$ [1]
 (c) It keeps expanding as it rises and eventually bursts, because the rubber can only stretch so far [1]

Accept any correct route in (b), including “the pressure is a quarter, so the volume is four times as big”. A student who multiplies instead of dividing gets $0.5\,m^3$ — a balloon that shrinks as it rises, which should have been rejected on sight.

- 3 (a) **X**, with **twice** the pressure of Y [1] [4]
 (b) Twice as many particles in the same volume, all moving at the same average speed since the temperature is the same [1], so twice as many collisions per unit area per second [1]
 (c) The kelvin temperature must double: **600 K** [1]

The trap in (c) is answering $600\,^{\circ}\text{C}$, or doubling $27\,^{\circ}\text{C}$. The question gives the temperature in kelvin precisely so that a student who converts unnecessarily exposes the habit.

- 4 (a) In the **middle** of the tube [1]; equal relative molecular masses at the same temperature means equal average speeds, so both gases travel the same distance in the same time [1] [4]
 (b) Any sensible reason, e.g. the denser gas would sink through the lighter one under gravity [1], so the gases would be mixed by bulk movement (convection) as well as by diffusion, and the result would no longer be caused by diffusion alone [1]

Part (b) is deliberately open. Award both marks for any answer that identifies bulk movement of the gas as a competing process and says why that spoils the test.

- 5 (a) $20\,000\,000 \times 0.0004$ [1] = **8000 N** [1] [4]
 (b) $100\,000 \times 0.0004 = \mathbf{40\,N}$ [1]; resultant = $8000 - 40 = \mathbf{7960\,N}$ **outwards** [1]

Here the atmospheric contribution is only half a per cent, which is worth saying out loud: knowing when a term matters is as useful as knowing it exists.

- 6** The air sealed in the bag is at about 100 000 Pa, the pressure at sea level **[1]** [4]
Higher up there is less air above, so atmospheric pressure outside the bag is lower **[1]**
The pressure inside is now greater than the pressure outside, so there is a resultant outward force on the bag **[1]**
The bag expands until the pressures balance, or until it bursts **[1]**

A student who says “the air inside expands because there is less pressure” has the right idea and one mark. Insist on the comparison between inside and outside for the rest.

- 7** (i) The mistake: squashing a gas does not speed the particles up, so no collision is harder than before — only a change in temperature does that **[1]** [4]

Correct: the particles have a shorter distance to travel between the walls, so they collide with them more often, giving more collisions per unit area per second **[1]**

(ii) The mistake: a vacuum cannot pull. A region of low pressure still pushes outwards, just far more weakly — it never draws anything towards it **[1]**

Correct: atmospheric pressure outside pushes the can inwards, and with the steam condensed there is nothing left inside to push back **[1]**